

Mathematics of Machine Learning

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Abstract

These notes cover the foundational mathematics of machine learning, focusing on statistical learning theory, empirical risk minimization, and popular algorithms such as decision trees, random forests, and neural networks.

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1 Introduction

Consider a pair of random variables $(X, Y) \in \mathcal{X} \times \mathcal{Y}$ with joint distribution P_0 , where X is to be thought of as an input or vector of predictors, and Y as an output or response. For instance X may represent a collection of disease risk factors (e.g. BMI, age, genetic indicators etc.) for a subject randomly selected from a population and Y may represent their disease status; or X could represent the number of bedrooms and other facilities in a randomly selected house, and Y could be its price.

In the former case we may take $\mathcal{Y} = \{-1, 1\}$ and this setting is known as the (two-class) classification setting. The latter case where $Y \in \mathbb{R}$ is an instance of a regression setting. We will take $\mathcal{X} = \mathbb{R}^p$ unless otherwise specified. We refer to Y as the output, or response, and X as the input and its components as predictors or variables.

It is of interest to predict the random Y from X ; we may attempt to do this via a (measurable) function $h : \mathcal{X} \rightarrow \mathcal{Y}$ known in the machine learning literature as a hypothesis. To measure the quality of such a prediction we will introduce a loss function $l : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$.

In the classification setting, loss given by the misclassification error is particularly relevant:

$$l(h(x), y) = \begin{cases} 0 & \text{if } h(x) = y \\ 1 & \text{otherwise.} \end{cases} \quad (1)$$

In this context h is also referred to as a classifier. In regression settings, the use of squared error $l(h(x), y) = (h(x) - y)^2$ is common, and we will take this to be the case unless specified otherwise. We will aim to pick a hypothesis h such that the risk

$$R(h) := \int_{(x,y) \in \mathcal{X} \times \mathcal{Y}} l(h(x), y) dP_0(x, y) \quad (2)$$

is small. For a deterministic h , $R(h) = \mathbb{E}l(h(X), Y)$.

Recall that the function h that minimises the risk in a regression setting is $\mathbb{E}(Y|X = x)$, which we refer to as the regression function. A classifier h_0 that minimises the misclassification risk is known as a Bayes classifier, and its risk is called the Bayes risk. A key function in the classification context is $\eta(x) := \mathbb{P}(Y = 1|X = x)$ which is also known as the regression function here.

Proposition 1.1. *A Bayes classifier h_0 is given by*

$$h_0(x) = \begin{cases} 1 & \text{if } \eta(x) > 1/2 \\ -1 & \text{otherwise.} \end{cases} \quad (3)$$

(When $\eta(x) = 1/2$ we can equally well take $h_0(x) = \pm 1$ and achieve the same misclassification error.)

In most settings of interest, the joint distribution P_0 of (X, Y) will be unknown. Instead we will suppose we have i.i.d. copies $(X_1, Y_1), \dots, (X_n, Y_n)$ known as training data. Our task is to use this data to construct a classifier $\hat{h}$ such that $R(\hat{h})$ or $\mathbb{E}R(\hat{h})$ is small.

Important point: $R(\hat{h})$ is a random variable depending on the random training data:

$$R(\hat{h}) = \mathbb{E}(l(\hat{h}(X), Y)|X_1, Y_1, \dots, X_n, Y_n). \quad (4)$$

1.1 Brief review of conditional expectation

For many of the mathematical arguments in this course we will need to manipulate conditional expectations. Recall that if $Z \in \mathbb{R}$ and $W = (W_1, \dots, W_d)^\top \in \mathbb{R}^d$ are random variables with

joint probability density function (pdf) $f_{Z,W}$ with respect to measure μ , then the conditional pdf $f_{Z|W}$ of Z given W satisfies

$$f_{Z|W}(z|w) = \begin{cases} f_{Z,W}(z, w)/f_W(w) & \text{if } f_W(w) \neq 0 \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

where f_W is the marginal pdf of W . Suppose $\mathbb{E}|Z| < \infty$. Then the conditional expectation function $\mathbb{E}(Z|W = w)$ is given by

$$g(w) := \mathbb{E}(Z|W = w) = \int z f_{Z|W}(z|w) \mu(dz). \quad (6)$$

We will make frequent use of the following properties:

1. **Role of independence:** If Z and W are independent, then $\mathbb{E}(Z|W) = \mathbb{E}Z$. If additionally for a random variable U , W is independent of (Z, U) , then $\mathbb{E}(Z|U, W) = \mathbb{E}(Z|U)$.
2. **Tower property:** Let $f : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be a (measurable) function. Then $\mathbb{E}\{\mathbb{E}(Z|W)|f(W)\} = \mathbb{E}\{Z|f(W)\}$.
3. **Fixing what is known:**

$$\mathbb{E}\{f(W_1, \dots, W_d)|W_1 = w_1, \dots, W_r = w_r\} = \mathbb{E}\{f(w_1, \dots, w_r, W_{r+1}, \dots, W_d)|W_1 = w_1, \dots, W_r = w_r\}$$

provided the r.h.s. is well-defined. In particular, if $\mathbb{E}Z^2 < \infty$ and $g : \mathbb{R}^d \rightarrow \mathbb{R}$ is such that $\mathbb{E}\{[g(W)]^2\} < \infty$ then $\mathbb{E}\{g(W)Z|W\} = g(W)\mathbb{E}(Z|W)$.

4. **Best least squares predictor:** With the conditions in (iii) above, we have $\mathbb{E}\{Z - g(W)\}^2 = \mathbb{E}\{Z - \mathbb{E}(Z|W)\}^2 + \mathbb{E}\{\mathbb{E}(Z|W) - g(W)\}^2$.

1.2 Bayes risk

Proof of Proposition 1. We have $R(h) = \mathbb{E}\mathbb{I}_{\{Y \neq h(X)\}} = \mathbb{E}\mathbb{E}[\mathbb{I}_{\{Y \neq h(X)\}}|X]$. Now

$$\begin{aligned} \mathbb{E}[\mathbb{I}_{\{Y \neq h(X)\}}|X] &= \mathbb{E}[\mathbb{I}_{\{Y=1\}}\mathbb{I}_{\{h(X)=-1\}} + \mathbb{I}_{\{Y=-1\}}\mathbb{I}_{\{h(X)=1\}}|X] \\ &= \mathbb{I}_{\{h(X)=-1\}}\eta(X) + \mathbb{I}_{\{h(X)=1\}}(1 - \eta(X)). \end{aligned}$$

When $\eta(X) > 1 - \eta(x)$ and so $\eta(X) > 1/2$, this is minimised by taking $h(X) = 1$, and similarly when $\eta(X) < 1/2$ this is minimised by taking $h(X) = -1$. $\square$

1.3 Empirical risk minimisation

Empirical risk minimisation replaces the expectation over the unknown P_0 in the definition of the risk with the empirical distribution, and seeks to minimise the resulting objective over $h \in \mathcal{H}$:

$$\hat{R}(h) := \frac{1}{n} \sum_{i=1}^n l(h(X_i), Y_i) \quad \text{and} \quad \hat{h} \in \arg \min_{h \in \mathcal{H}} \hat{R}(h). \quad (7)$$

$\hat{R}(h)$ is the empirical risk or training error of h and $\hat{h}$ is the empirical risk minimiser (ERM).

Consider the regression setting with squared error loss where $\mathcal{H} = \{x \mapsto \mu + x^\top \beta\}$ for $\mu \in \mathbb{R}, \beta \in \mathbb{R}^p$. Then empirical risk minimisation is equivalent to ordinary least squares, giving $\hat{h}(x) = \hat{\mu} + \hat{\beta}^\top x$. Forming matrix $\Phi \in \mathbb{R}^{n \times d}$ with entries $\Phi_{ij} = \varphi_j(X_i)$ assumed to be of full column rank, we have that the ERM $\hat{\beta} = (\Phi^\top \Phi)^{-1} \Phi^\top Y_{1:n}$.

1.4 Bias-variance tradeoff

Let us consider $\hat{h} = \hat{h}_D$ trained on data $D = (X_i, Y_i)_{i=1}^n$ formed of iid copies of an independent random pair (X, Y) . Taking expectations over the conditional risk:

$$\mathbb{E}R(\hat{h}_D) = \mathbb{E}\{\mathbb{E}(Y|X) - \bar{h}(X)\}^2 + \mathbb{E}\text{Var}(\hat{h}_D(X)|X) + \mathbb{E}\text{Var}(Y|X) \quad (8)$$

where the first term is squared bias, the second is variance of $\hat{h}$, and the last is irreducible variance. If $\hat{h}$ were an ERM over class $\mathcal{H}$, a rich class of hypotheses would result in a smaller squared bias term, but the variance would likely increase as it fits the training data D more closely.

1.5 Cross-validation

The idea of v -fold cross-validation is to split the data into v groups or folds of roughly equal size. Let D_{-k} be all the data except that in the k th fold, and let $A_k \subset \{1, \dots, n\}$ be the observation indices corresponding to the k th fold. We choose the value of j that minimises

$$CV(j) := \frac{1}{n} \sum_{k=1}^v \sum_{i \in A_k} l(\hat{h}_{-k}^{(j)}(X_i), Y_i). \quad (9)$$

where $\hat{h}_{-k}^{(j)}$ is the hypothesis obtained applying method j on D_{-k} .

2 Popular machine learning methods I

2.1 Decision trees

Regression trees use a set of basis functions consisting of indicator functions on rectangular regions and take the form:

$$T(x) = \sum_{j=1}^J \gamma_j \mathbb{I}_{R_j}(x); \quad (10)$$

here R_j are rectangular regions that form a partition of $\mathbb{R}^p$ and γ_j are coefficients in $\mathbb{R}$.

The regions and coefficients are typically computed from data using a recursive binary partitioning algorithm. When aiming to minimise the overall residual sum of squares (RSS), we search for a predictor $\hat{f}_R$ and split point $\hat{s}_R$ that minimise the RSS on the left and right splits.

To compute splits efficiently for sorted data $X_1 < X_2 < \dots < X_n$, we can equivalently find m to minimise $Q_m + P_m$ where Q_m is the RSS for $i \leq m$ and P_m is the RSS for $i > m$. As it can be shown that $P_m + Q_m = \sum_{i=1}^n Y_i^2 - \frac{1}{m} (\sum_{i \leq m} Y_i)^2 - \frac{1}{n-m} (\sum_{i > m} Y_i)^2$, these terms can be updated in $O(n)$ operations.

2.2 Random forests

While decision trees are useful, they have high variance over the training data and produce piecewise constant functions. Random forest effectively attempts to 'estimate' $\bar{T}$ and so improve upon the variance of a single tree. Random forest samples the data D with replacement to form new datasets $D_1^*, \dots, D_B^*$ and grows a decision tree $\hat{T}^{(b)}$ for each, but randomly samples m_{try} of the p predictors when choosing splits. The output is $f_{rf} = \frac{1}{B} \sum_{b=1}^B \hat{T}^{(b)}$.

The reason for sampling predictors is to make the trees more independent. If the trees have correlation $\rho \geq 0$, the variance of the ensemble is

$$\text{Var}(f_{rf}(x)) = \frac{1-\rho}{B} \text{Var}(\hat{T}^{(1)}(x)) + \rho \text{Var}(\hat{T}^{(1)}(x)). \quad (11)$$

While the first term shrinks with large B , the second does not depend on B , so we want ρ to be small, which the extra randomisation helps to achieve.

3 Statistical learning theory

Our goal now is to study a roughly analogous quantity in the classification setting. For an ERM $\hat{h}$ over a class $\mathcal{H}$, we may compare the risk of $\hat{h}$ to the best hypothesis $h^* := \arg \min_{h \in \mathcal{H}} R(h)$. The quantity $R(\hat{h}) - R(h^*)$ is known as the excess risk.

3.1 Sub-Gaussianity and Hoeffding's inequality

Definition 3.1. We say a random variable W is sub-Gaussian with parameter $\sigma > 0$ if

$$\mathbb{E} e^{\alpha(W - \mathbb{E}W)} \leq e^{\alpha^2 \sigma^2 / 2} \quad \text{for all } \alpha \in \mathbb{R}. \quad (12)$$

Lemma 3.2 (Hoeffding's lemma). *If W takes values in $[a, b]$, then W is sub-Gaussian with parameter $\sigma = (b - a)/2$.*

If $W_1, \dots, W_n$ are independent and each $a_i \leq W_i \leq b_i$ almost surely, Hoeffding's inequality states:

$$\mathbb{P} \left(\frac{1}{n} \sum_{i=1}^n (W_i - \mathbb{E}W_i) \geq t \right) \leq \exp \left(- \frac{2n^2 t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right) \quad \text{for } t \geq 0. \quad (13)$$

For finite hypothesis classes, if $\mathcal{H}$ is finite, then with probability at least $1 - \delta$ the ERM $\hat{h}$ satisfies $R(\hat{h}) - R(h^*) \leq \sqrt{\frac{2(\log |\mathcal{H}| + \log(1/\delta))}{n}}$.

3.2 Rademacher complexity

Definition 3.3. Let $\mathcal{F}$ be a class of real-valued functions $f : \mathcal{Z} \rightarrow \mathbb{R}$ and let $z_1, \dots, z_n \in \mathcal{Z}$. Define the empirical Rademacher complexity:

$$\hat{\mathcal{R}}(\mathcal{F}(z_{1:n})) := \mathbb{E} \left(\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \epsilon_i f(z_i) \right) \quad (14)$$

where $\epsilon_1, \dots, \epsilon_n$ are i.i.d. Rademacher random variables.

Theorem 3.4. *Let $\mathcal{F}$ be a class of real-valued functions $f : \mathcal{Z} \rightarrow \mathbb{R}$ and let $Z_1, \dots, Z_n$ be i.i.d. random variables taking values in $\mathcal{Z}$. Then*

$$\mathbb{E} \left(\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \{f(Z_i) - \mathbb{E}f(Z_i)\} \right) \leq 2\mathcal{R}_n(\mathcal{F}). \quad (15)$$

3.3 VC dimension

Definition 3.5. Let $\mathcal{H}$ be a class of functions $h : \mathcal{X} \rightarrow \{a, b\}$ with $a \neq b$. The VC dimension $VC(\mathcal{H})$ is the largest integer n such that some $x_{1:n}$ is shattered by $\mathcal{H}$. Equivalently, $VC(\mathcal{H}) = \sup\{n \in \mathbb{N} : s(\mathcal{H}, n) = 2^n\}$.

Lemma 3.6 (Sauer-Shelah). *Let $\mathcal{H}$ be a class with finite VC dimension d . Then $s(\mathcal{H}, n) \leq (n+1)^d$.*

The Sauer-Shelah lemma is remarkable because it shows that beyond d , the growth of the shattering coefficient $s(\mathcal{H}, n)$ changes radically to become polynomial. Consequently, the Rademacher complexity is bounded by $\mathcal{R}_n(\mathcal{F}) \leq \sqrt{\frac{2VC(\mathcal{H}) \log(n+1)}{n}}$.

4 Computation for empirical risk minimisation

4.1 Convex surrogates

In the classification setting, one problem with using misclassification loss is that the ERM optimisation can be intractable for many hypothesis classes. For example, the indicator function $\mathbb{I}_{\{-\infty, 0\}}$ is not convex and not continuous. We will consider loss functions $l : \mathbb{R} \times \mathbb{R} \rightarrow [0, \infty)$ of the form $l(h(x), y) = \phi(yh(x))$ where $\phi : \mathbb{R} \rightarrow [0, \infty)$ is convex. Common choices of ϕ include:

- Hinge loss: $\phi(u) = \max(1 - u, 0)$.
- Exponential loss: $\phi(u) = e^{-u}$.
- Logistic loss: $\phi(u) = \log_2(1 + e^{-u})$.

4.2 Constraint Classes and Rademacher Bounds

To keep Rademacher complexity finite in vector spaces, we often constrain norms.

- **l_2 -constraint:** For $\mathcal{H} = \{x \mapsto x^\top \beta : \|\beta\|_2 \leq \lambda\}$, the Rademacher complexity is bounded by $\mathcal{R}_n(\mathcal{H}) \leq \frac{\lambda(\mathbb{E}\|X\|_2^2)^{1/2}}{\sqrt{n}}$.
- **l_1 -constraint:** For $\|\beta\|_1 \leq \lambda$ and $\mathcal{X} = [-C, C]^p$, the complexity is bounded by $\hat{\mathcal{R}}(\mathcal{H}(x_{1:n})) \leq \frac{\lambda C}{\sqrt{n}} \sqrt{2 \log(2p)}$.

If every predictor is equally important, the l_2 constraint tends to perform better. If only s predictors are important and s is small, the l_1 approach can perform well.

4.3 Projections and Subgradients

Let $C \subseteq \mathbb{R}^d$ be a closed convex set. For each $x \in \mathbb{R}^d$, the minimiser of $\|x - z\|_2$ over $z \in C$ exists and is unique, denoted as the projection $\pi_C(x)$. We have $(x - \pi_C(x))^\top (z - \pi_C(x)) \leq 0$ for all $z \in C$.

A vector $g \in \mathbb{R}^d$ is a subgradient of a convex function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ at x if $f(z) \geq f(x) + g^\top (z - x)$ for all $z \in \mathbb{R}^d$.

4.4 Gradient descent and Stochastic Gradient Descent (SGD)

The procedure of (projected) gradient descent for minimising f over a closed convex set C uses this intuition to produce a sequence of iterates $\beta_1, \beta_2, \dots$ aiming to have $f(\beta_s)$ close to a minimum $f(\hat{\beta})$ for large s . Stochastic gradient descent can circumvent the issue of computationally expensive gradients by evaluating the gradient on a single data point (x_{U_s}, y_{U_s}) at each step. With an appropriate step size, the stochastic gradient descent algorithm satisfies $\mathbb{E}f(\bar{\beta}) - f(\hat{\beta}) \leq \frac{2LR}{\sqrt{k}}$.

5 Popular machine learning methods II

5.1 Adaboost

Given a tuning parameter M , Adaboost performs greedy empirical risk minimisation over a weighted combination of base classifiers using exponential loss. The algorithm computes weights $w_i^{(m)} = n^{-1} \exp(-Y_i \hat{f}_{m-1}(X_i))$ and selects h_m to minimise the weighted error $\text{err}_m(h)$. The optimal weight for the classifier is $\hat{\beta}_m = \frac{1}{2} \log \left(\frac{1 - \text{err}_m(\hat{h}_m)}{\text{err}_m(\hat{h}_m)} \right)$.

5.2 Gradient Boosting

Gradient boosting aims to minimise an empirical version of the risk by using regression to approximate the conditional expectation. The algorithm repeatedly computes $W_i = \frac{\partial}{\partial \theta} l(\theta, Y_i) |_{\theta = \hat{f}_{m-1}(X_i)}$, which in the case of least squares loss is $-2(Y_i - \hat{f}_{m-1}(X_i))$, and then regresses these negative residuals back onto $X_{1:n}$ to update the model $\hat{f}_m = \hat{f}_{m-1} - \eta \hat{g}_m$.

5.3 Feedforward neural networks

The class of feedforward neural networks are based around a particular class of hypotheses $h : \mathcal{X} = \mathbb{R}^p \rightarrow \mathbb{R}$ with general form:

$$h(x) = A^{(d)} \circ g \circ A^{(d-1)} \circ g \circ \dots \circ g \circ A^{(2)} \circ g \circ A^{(1)}(x) \quad (16)$$

where d is known as the depth of the network, $A^{(k)}(v) = \beta^{(k)}v + \mu^{(k)}$, and g applies a nonlinear activation function elementwise. Typical choices include the rectified linear unit (ReLU) $u \mapsto \max(u, 0)$ and the sigmoid $u \mapsto 1/(1 + e^{-u})$.

Parameters $(\beta^{(k)}, \mu^{(k)})_{k=1}^d$ are typically fitted using empirical risk minimisation via stochastic gradient descent. A key factor in their success is that the gradients can be computed quickly using the chain rule via backpropagation. For a loss $z = \phi(yh(x))$, we iteratively compute intermediate terms such as $\frac{\partial z}{\partial x^{(d)}} = y\phi'(yx^{(d)})$ down through the hidden layers.